

YAKUSHIMA AP, Japan

WMO: 478360

Lat: 30.386N

Lon: 130.659E

Elev: 38

StdP: 100.87

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.9	6.9	-1.8	3.3	9.4	-0.7	3.6	9.5	15.2	10.5	14.2	10.2	6.7	290	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.5	31.6	26.3	30.9	26.2	30.3	25.9	27.3	29.9	26.9	29.5	26.5	29.2	4.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.4	28.8	26.2	21.7	28.6	25.7	21.1	28.4	87.0	30.0	85.0	29.5	83.2	29.3	31.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.6	12.0	10.5	3.9	33.1	1.1	0.9	3.1	33.8	2.4	34.4	1.8	34.9	1.0	35.6		
(5)			1.9	28.0	0.8	0.9	1.3	28.7	0.9	29.2	0.4	29.7	-0.2	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	11.6	12.3	14.4	17.7	21.0	23.8	27.1	27.6	25.8	22.4	18.1	13.8	(6)	
(7)		DBStd	6.00	2.72	3.03	2.77	2.35	1.78	2.11	1.28	1.02	1.62	2.15	2.65	2.93	(7)	
(8)		HDD10.0	28	13	9	2	0	0	0	0	0	0	0	0	3	(8)	
(9)		HDD18.3	721	208	169	124	39	2	0	0	0	0	1	36	143	(9)	
(10)		CDD10.0	3563	63	74	139	231	341	413	531	547	475	384	244	121	(10)	
(11)		CDD18.3	1217	1	1	3	20	84	163	273	289	225	127	29	2	(11)	
(12)		CDH23.3	9438	0	0	3	17	198	886	2822	3135	1891	459	28	1	(12)	
(13)		CDH26.7	2457	0	0	0	0	7	153	869	1011	387	29	1	0	(13)	
(14)	Wind	WSAvg	4.6	5.9	5.8	5.5	4.8	4.1	3.3	3.2	3.7	4.1	4.7	4.9	5.7	(14)	
(15)	Precipitation	PrecAvg	4267	269	259	345	380	412	795	327	239	407	291	292	252	(15)	
(16)		PrecMax	5881	573	660	732	712	813	1251	900	550	1141	758	572	601	(16)	
(17)		PrecMin	2101	17	94	98	160	113	324	10	37	80	28	123	51	(17)	
(18)		PrecStd	659	117	123	139	152	188	253	237	148	265	190	130	114	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.5	21.6	23.1	25.1	27.6	31.4	32.6	32.7	31.2	28.7	25.7	22.1	(19)	
(20)		MCWB	17.0	18.2	18.0	20.2	22.2	25.6	25.8	26.5	26.2	24.3	22.1	18.2	(20)		
(21)		2%	DB	18.7	20.0	21.2	23.6	26.1	29.8	31.6	31.7	30.2	27.5	24.0	20.5	(21)	
(22)		MCWB	15.5	17.2	17.1	19.7	20.8	25.2	26.2	26.5	25.7	23.8	20.3	17.0	(22)		
(23)		5%	DB	17.3	18.6	20.0	22.6	25.2	28.5	30.9	31.0	29.5	26.5	23.0	19.4	(23)	
(24)		MCWB	14.3	15.8	16.5	18.9	20.6	24.9	26.1	26.4	25.3	22.9	19.4	16.2	(24)		
(25)		10%	DB	16.0	17.2	18.9	21.7	24.4	27.1	30.2	30.4	28.8	25.6	22.1	18.4	(25)	
(26)		MCWB	13.0	14.4	15.6	18.1	20.3	24.3	25.9	26.1	24.9	22.1	18.7	15.2	(26)		
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	19.3	20.1	22.6	24.4	26.6	27.9	28.1	27.3	26.0	23.0	19.6	(27)
(28)			MADB	19.7	20.8	21.5	23.6	26.1	29.3	30.3	30.3	29.7	27.2	24.7	20.7	(28)	
(29)	2%		WB	16.3	17.7	18.7	21.4	23.5	25.8	27.2	27.4	26.6	24.8	21.5	18.1	(29)	
(30)	MADB		18.0	19.3	20.2	22.6	24.9	28.3	29.9	29.9	28.9	26.5	23.2	19.8	(30)		
(31)	5%		WB	14.9	16.3	17.5	20.2	22.6	25.3	26.8	27.0	26.1	23.9	20.4	17.0	(31)	
(32)	MADB		17.0	18.1	19.2	21.6	24.0	27.6	29.5	29.7	28.3	25.8	22.2	18.9	(32)		
(33)	10%		WB	13.3	14.9	16.3	19.1	21.7	24.7	26.4	26.7	25.7	23.0	19.5	15.7	(33)	
(34)	MADB		15.7	16.8	18.2	20.8	23.2	26.8	29.2	29.3	27.9	25.0	21.5	18.0	(34)		
(35)	Mean Daily Temperature Range		MDBR	4.9	5.3	5.9	6.3	6.2	5.0	5.7	5.5	5.3	5.0	5.2	5.0	(35)	
(36)			5% DB	MCDBR	7.2	7.1	7.4	7.4	7.3	6.4	6.8	6.4	6.1	5.8	5.8	6.1	(36)
(37)		MCWBR	6.0	6.0	5.9	5.4	5.1	3.5	3.7	3.5	3.5	3.7	4.3	4.9	(37)		
(38)		5% WB	MCDDBR	6.5	6.4	6.1	5.6	5.3	5.6	6.0	5.6	5.3	4.7	4.9	5.0	(38)	
(39)		MCWBR	6.2	6.3	6.1	5.2	4.7	3.4	3.6	3.2	3.3	3.4	4.4	4.9	(39)		
(40)	Clear-Sky Solar Irradiance	taub	0.444	0.476	0.528	0.540	0.531	0.523	0.500	0.494	0.486	0.474	0.444	0.437	(40)		
(41)		taud	2.094	2.039	1.911	1.939	2.034	2.149	2.242	2.259	2.239	2.185	2.193	2.113	(41)		
(42)		Ebn at Noon	760	775	763	772	780	782	799	800	793	770	758	743	(42)		
(43)		Edh at Noon	135	155	188	189	173	154	140	136	134	133	121	126	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.13	2.77	3.77	4.54	4.80	4.17	5.87	5.77	4.66	3.65	2.58	2.09	(44)		
(45)		RadStd	0.23	0.34	0.28	0.43	0.54	0.50	0.55	0.34	0.47	0.40	0.31	0.21	(45)		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.41	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page